Koushik Botcha

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Hyderabad, Telangana

Professional Summary

Full-stack Flutter Developer with hands-on experience building production-grade mobile applications across healthcare, sustainability, and security domains. Proficient in cross-platform and native Android development, real-time systems (WebSocket), AI integration (Google Gemini), biometric and NFC authentication, and backend engineering with Node.js and PostgreSQL. Proven ability to architect, ship, and maintain scalable consumer-facing products with CI/CD pipelines.

Experience

•TechOptima Solutions Pvt. Ltd. *Jan 2025 – Jun 2025*
Software Development Intern Hyderabad, Telangana

- Engineered 4+ production-grade mobile applications using Flutter, Kotlin, and Node.js spanning healthcare, sustainability, and security verticals
- Implemented secure multi-factor authentication combining biometrics, NFC card reading, and Aadhaar XML verification
- Architected RESTful APIs with Node.js/Express and PostgreSQL achieving 99.5% uptime across all services
- Integrated Google Gemini AI for intelligent symptom analysis and conversational healthcare assistance
- Built real-time GPS tracking with Mapbox and Google Maps API for ambulance and logistics routing
- Established CI/CD pipelines via GitHub Actions reducing deployment time by 60% with 85%+ test coverage

Projects

•MedTek – Digital Healthcare Ecosystem *Jan 2026 – Present*
Full-stack telemedicine with AI-powered triage, real-time ride logistics, and pill detection

- Tools: Flutter, Node.js, Express.js, PostgreSQL, Google Gemini AI, Mapbox, WebRTC, Speech-to-Text, TTS
- Built 27+ screen healthcare platform with role-based dashboards for patients, doctors, and admins
- Integrated Google Gemini AI for symptom triage with hands-free voice interface (Speech-to-Text + TTS)
- Developed ambulance ride-booking with real-time GPS tracking, PIN verification, and Haversine fare estimation
- Implemented PostgreSQL schema with 7 tables, JWT authentication, and bcrypt password hashing
- Created AI pill detection with image recognition, confidence scoring, and drug interaction cross-checks

•Swach – Waste Management & Carbon Trading Platform *Dec 2025 – Present*
Sustainability platform with waste pickup services, digital wallet, and carbon credit ecosystem

- Tools: Flutter, Node.js, Express.js, PostgreSQL, Sequelize ORM, Firebase (FCM, Auth), Provider, GetX
- Developed platform with 13 Node.js API modules and dual Flutter apps (Customer + Admin) spanning 47+ screens
- Built carbon marketplace and footprint calculator aligned to GHG Protocol Scope 1/2/3 emissions standards
- Implemented digital wallet with transaction history and automatic carbon credit allocation (1 credit/kg waste)
- Created live order tracking with Firebase Cloud Messaging notifications and driver assignment workflows
- Designed PostgreSQL schema with 12 entities using Sequelize ORM, secured with JWT authentication

•Airport Security System with NFC & Biometric Auth *Jul 2025 – Nov 2025*
Multi-platform identity verification with Aadhaar XML, NFC, and barcode scanning

- Tools: Flutter, Native Android (Kotlin), NFC Technology, Aadhaar XML API, Barcode Scanner, Biometric Auth
- Built dual-platform app in Native Android (Kotlin) and Flutter for airport security personnel verification
- Implemented NFC card reading and Aadhaar XML API integration for government-grade identity validation
- Developed barcode scanning to parse passenger and flight details for downstream security processing
- Designed dual-factor verification combining biometric auth (fingerprint + face ID) with document scanning

•DevPulse – Developer Analytics Dashboard *Feb 2026 – Present*
Real-time personal dev metrics with GitHub and LeetCode integration

- Tools: Flutter, Node.js, WebSocket (ws), PostgreSQL, GitHub GraphQL API, LeetCode GraphQL API, fl_chart, Provider
- Built developer productivity dashboard aggregating GitHub contributions, streaks, and LeetCode stats in real time
- Architected WebSocket server with Node.js for bidirectional low-latency data sync from multiple external APIs
- Implemented PostgreSQL caching layer with 30-minute cron refresh cycle ensuring instant app loads under 100ms
- Designed animated fl_chart visualizations for contribution graphs, solve rate trends, and streak heatmaps
- Created daily briefing notifications and goal-tracking system with offline-first shared preferences persistence

•Real-Time Chat & Collaboration Application *Feb 2026 – Present*
WebSocket-powered messaging with group chat, media sharing, and presence tracking

- Tools: Flutter, Node.js, WebSocket (ws library), PostgreSQL, Provider, Firebase Cloud Messaging
- Building real-time messaging using WebSocket for bidirectional low-latency communication
- Implementing group chat with typing indicators, read receipts, and delivery status tracking
- Developing media sharing for images, documents, and voice messages with cloud storage integration
- Creating user presence system showing online/offline status with last-seen timestamps

Education

•Shri Ramakrishna Raju Engineering College (SRKR), Bhimavaram *2020 – 2024*
Bachelor of Technology in Civil Engineering CGPA: 7.69/10.0

•Sri Chaitanya Junior College, Visakhapatnam *2018 – 2020*
Board of Intermediate Education, Andhra Pradesh Percentage: 81%

Technical Skills

Languages: Dart, Kotlin, Python, JavaScript (Node.js), SQL
Mobile Development: Flutter, Native Android (Kotlin), Material Design 3, Provider, GetX, Riverpod, BLoC
Backend & Databases: Node.js, Express.js, PostgreSQL, Sequelize ORM, Firebase (Firestore, Auth, FCM), REST APIs
Real-Time & AI: WebSocket, Socket.IO, Google Gemini AI, TensorFlow Lite, Speech-to-Text, TTS, WebRTC
Security & Auth: JWT, bcrypt, NFC, Biometric Auth, Aadhaar XML, API Key Validation
Maps & Location: Google Maps API, Mapbox, Geolocator, Geocoding, Route Optimization
Cloud & DevOps: Firebase Services, Cloudinary CDN, CI/CD (GitHub Actions), Docker
Developer Tools: Git, GitHub, Android Studio, VS Code, Postman, Flutter Test, Mockito

Certifications & Achievements

- **Flutter & Dart – Complete Guide**– Udemy, Maximilian Schwarzmüller *2024*
- **Google Associate Android Developer**– Google Developers | Credential: GAD-KOUSHIK-2024 *2024*
- **AWS Certified Cloud Practitioner**– Amazon Web Services | Validation: AWS-CCP-KB-2024 *2024*
- **T-Hub Hackathon by Wincible – 5th Position**– Healthcare solution built in 24-hour competitive hackathon (Team Lead) *2024*
- **Smart India Hackathon – Finalist**– Top 50 teams nationwide from 10,000+ participants *2024*
- **Best Final Year Project Award**– AI Civil Infrastructure System, Civil Engineering Dept. *2024*

Hobbies & Interests

- **Team Sports & Leadership**– Active basketball player in inter-college tournaments; demonstrated team coordination and leadership
- **Hackathon Enthusiast**– Regular participant in competitive coding events and 24-hour hackathons
- **Open Source & Community**– Contributing to open-source Flutter projects and mentoring junior developers

Areas of Interest

Cross-Platform Development, AI/ML Integration in Mobile Apps, Real-Time Systems & WebSocket Architecture, Developer Tooling & Analytics, NFC & Biometric Security, Carbon Trading & Sustainability Tech, Geospatial Applications, Voice Interfaces, Full-Stack Mobile Development, Cloud-Native Applications